Balaji Arjun M R L

Embedded Software Engineer

Personal Info

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LinkedIn

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Skills

Programming in C, C++

Linux

CAN

Autosar: Configuration, MCAL,

Unit testing, V&V,

SPI, I2C, UART

Python,

Infineon Aurix TC392X

Tools & Compilers

Trace32

AURIX Development Studio

Eclipse IDE

Vector CAST, Davinci

Gtest , Visual Studio

C4K

WinSCP

PuTTY

Languages

English

Hindi

Tamil

I am **3.6** years of experienced professional in Embedded Systems industry, with **excellent programming** skills coupled **in-depth µC knowledge**. I have an exceptional ability to **develop and troubleshoot** software, combined with good problem solving and communication skills. My background is in Autosar, real-time embedded programming, communication protocols and other automotive systems. Looking for an challenging opportunity to bring the ***ideas into action***!

Education

2015 - 2018

Completed bachelor's degree in **Electrical & Electronics** from **Thiagrajar College of Engineering** Madurai with **72%**, also scored **83.8%** and **95.31%** marks in SSLC and Diploma (EEE).

Experience

2021-11 - present

Autosar Software Engineer

KPIT Technologies Pvt Ltd,. Bangalore

1. HONDA CORE ECU

- Project Type : Adaptive Autosar Development in Telematics & DAQ Services.
- Technology : **Adaptive Autosar**, C++, **CAN**, **DAQ** , Unit testing, Python.
- Tools Used : C4K, WinSCP, PuTTY, Gtest.
- Platform : Telematics and DAQ Services.

** This project includes **developing and testing** the Adaptive Autosar (**KSAR** stack)part in the Telematics and DAQ services.*
** In this verifying and validation (V & V) in the wipe personal data service and DAQ services. Creating test cases for the requirements.*
** Configuring the application by using C4K tool. Deploying and Running the application in Qemus using Linux and verifying the logs.*

2. QUALCOMM QRIDE

- Project Type : Adaptive Autosar development in functional clusters.
- Technology : **Adaptive Autosar**, Embedded C++.
- Tools Used : VectorDavinci, WinSCP, PuTTY.
- Platform : Default services and Diagnostic manager.

** This project includes **developing and testing** the Adaptive Autosar (Vector stack) part in the vector default services,*
** Sample Application to simulate faults from DOIP, Reading and writing of examples diagnostics parameters using UDS commands,*
** created test cases for the requirements.Configured the application by using VectorDavinci Tool, Builded in QNX,Deployed and run the application in Linux and verified the logs.*

3. ZF IP NEXT

- Project Type : Classic Autosar development using AURIX Board.
- Technology : **Classic Autosar**, Embedded C & C++.
- Tools Used : Eclipse, AURIX Development Studio, Trace32, VectorCAST.
- Platform : Aurix TC392X.

** During this project **Developed and tested** the Classic Autosar (Infineon Stack) Part in the SPI Loopback Communication by using Aurix Board (TC392X),*
** Implemented the service component for QNX to run on the ISOC of UART Communication,Test Cases for the Software Integration test.*
** Unit test cases for the QNX to run on the ISOC of UART Communication by using VectorCAST tool.*

Interests

Automotive

AUTOSAR

Machine Learning

Artificial Intelligence

Firmware

Device Driver

Automation

Aerospace

Multimedia

Elecrical Energy

2021-02 -
2021-10

Embedded Design Engineer

Embien Technologies Pvt Ltd,. Madurai.

1. EMBIEN AI/ML

- Project Type : Research & Development of Artificial Intelligence & Machine Learning.
- Technology : AI/ML, Image Processing, Python, C++
- Tools Used : Jupyter, Google Colab, Microsoft Visual Studio.
- Platform : Camera Integration & Running application in CPU & GPU

** During this project **Developed and tested** the Implementation of AI/ML in Embedded Applications running the model in CPU,*
** GPU(NVIDIA) and Google Colab's CPU, GPU(Tesla K80) integrated with the webcam.*

2020-07 -
2021-01

Embedded Member Technical Engineer

Traana Technologies Pvt Ltd,. Bangalore.

1. GPRS (Generic Programmable Radar Simulator)

* Project Type : Graphical User Interface (GUI) of Radar Movements.
* Technology : Digital Signal Processing, C++, Socket Programming.
* Tools Used : Qt IDE
* Platform : Radar Integration and signal Processing
** During this project **Developed and tested** for radar integration with GUI in QT(MCPU) and Radar controller (RC). Performed the different radar Movements Movements in GUI servers and integrate with the effects of new addition of signal processing with the radar controller. These communications was performed by the socket programming technique and successfully completed the testing.*

2019-03 -
2019-06

Embedded Software Engineer Intern

CSK Electronics and automation Pvt Ltd,. Coimbatore.

1. ECOAIR AUTOMATION

* Technology : Embedded C, STM Series controller.
* During this project ***Developed and tested*** for Ecoair Cooling System integration with STM series controller and worked with the communication protocols. (I2C, SPI and UART). This automation performance was successfully tested.

Certifications

- Embedded Systems Course completed in **Vector India Pvt Ltd,.** - (Sep - 2019 to Apr - 2020)
- Python Development Course from Udemy
- Autosar Development Course from KPIT - E-Code - (Nov - 2021 to Aug - 2022)